

Assignments

T1 – Task One

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1 Issue 1 - Replace FAT by SPIFFS

In the basic setup, the FAT file system is used on both partitions. Replace the existing FAT file system configuration by a SPIFFS file system configuration and re-run the provided file system tests. The QEMU output should look similar to the test output provided below:

```
ELF-loader started on CPU: ARM Ltd. Cortex-A9 r3p0
  paddr=[717000..adeeb3]
No DTB passed in from boot loader.
Looking for DTB in CPIO archive...found at 8260bc.
Loaded DTB from 8260bc.
  paddr=[3c000..3efff]
ELF-loading image 'kernel'
  paddr=[0..3bfff]
  vaddr=[e0000000..e003bfff]
  virt_entry=e0000000
ELF-loading image 'capdl-loader'
  paddr=[3f000..317fff]
  vaddr=[10000..2e8fff]
  virt_entry=18824
Enabling MMU and paging
Bootstrapping kernel
Warning: Could not infer GIC interrupt target ID, assuming 0.
Booting all finished, dropped to user space
main@main.c:2114 Starting CapDL Loader...
main@main.c:2116 CapDL Loader done, suspending...
  INFO: /host/sdk/components/RamDisk/RamDisk.c:278: RamDisk has size of 2097152 bytes
  INFO: /host/sdk/components/StorageServer/server/src/storageServer_rpc.c:504: storage
medium size: 2097152 bytes
  INFO: /host/sdk/components/StorageServer/server/src/storageServer_rpc.c:522: client 1:
offset=0, size=1048576
  INFO: /host/sdk/components/StorageServer/server/src/storageServer_rpc.c:522: client 2:
offset=1048576, size=1048576
  INFO: /host/sdk/libs/os_filesystem/src/OS_FileSystem.c:125: Maximizing file system ac-
cording to size reported by the storage layer (1048576 bytes)
  INFO: /host/sdk/libs/os_filesystem/src/lib/SpifFs.c:182: Using SPIFFS (phys_addr = 0,
phys_size = 1048576, phys_erase_block = 4096, log_block_size = 4096, log_page_size = 256)
  INFO: /host/sdk/libs/os_filesystem/src/OS_FileSystem.c:125: Maximizing file system ac-
cording to size reported by the storage layer (1048576 bytes)
  INFO: /host/sdk/libs/os_filesystem/src/lib/SpifFs.c:182: Using SPIFFS (phys_addr = 0,
phys_size = 1048576, phys_erase_block = 4096, log_block_size = 4096, log_page_size = 256)
  INFO: /host/sdk/demos/demo_training_course/src/components/TestApp/src/TestApp.c:135:
Demo completed successfully.
```

Code Block 1: QEMU Terminal

2 Issue 2 - Build and run the demo on the RPi 3 B+

As a next step, we want to leave QEMU and execute our demo on the RPi 3 B+. Therefore, we have to do several modifications:

1. Adapt the build process of the basic setup in order to build for the RPi 3 B+ platform. As a source of inspiration, please have a look at the documentation of the RPi 3 B+ IoT demo, which you can find in the TRENTOS-M Handbook.
2. Prepare the SD card with all the required files: the RPi firmware blobs and the TRENTOS system image you built before. Again, have a look at the the documentation of the RPi 3 B+ IoT demo, in order to identify the right files. Hint: have a look at the folders `/sdk/resources` and the subfolder `images` of your RPi 3 B+ build folder.
3. Prepare the wiring of the RPi 3 B+ according to the setup mentioned by the documentation of the RPi 3 B+ IoT demo. For now, this only includes
 - a. electrical power
 - b. USB-to-UART adapter for connecting the RPi 3 B+ to the PC
4. Use a tool like `picocom` (see documentation of the RPi 3 B+ IoT demo) to receive terminal output from the RPi 3 B+.
5. Power on the RPi 3 B+.
6. You should receive an output similar like the one below:

```
U-Boot 2020.07-rc3-00002-g4b6f96a0a6 (May 26 2020 - 18:38:13 +0200)

DRAM:  948 MiB
RPI 3 Model B+ (0xa020d3)
MMC:   mmc@7e202000: 0, sdhci@7e300000: 1
Loading Environment from FAT... OK
In:    serial
Out:   vidconsole
Err:   vidconsole
Net:   No ethernet found.
starting USB...
Bus usb@7e980000: USB DWC2
scanning bus usb@7e980000 for devices... 4 USB Device(s) found
    scanning usb for storage devices... 0 Storage Device(s) found
Hit any key to stop autoboot:  0
4068580 bytes read in 171 ms (22.7 MiB/s)
## Starting application at 0x0070d000 ...

ELF-loader started on CPU: ARM Ltd. Cortex-A53 r0p4
  paddr=[70d000..ac96c7]
No DTB passed in from boot loader.
Looking for DTB in CPIO archive...found at 81c294.
Loaded DTB from 81c294.
```

Issue 2 - Build and run the demo on the RPi 3 B+

```
paddr=[3c000..3ffff]
ELF-loading image 'kernel'
paddr=[0..3bfff]
vaddr=[e0000000..e003bfff]
virt_entry=e0000000
ELF-loading image 'capdl-loader'
paddr=[40000..318fff]
vaddr=[10000..2e8fff]
virt_entry=1860c
Enabling MMU and paging
Jumping to kernel-image entry point...

Bootstrapping kernel
Booting all finished, dropped to user space
main@main.c:2114 Starting CapDL Loader...
main@main.c:2116 CapDL Loader done, suspending...
INFO: /host/sdk/components/RamDisk/RamDisk.c:278: RamDisk has size of 2097152 bytes
INFO: /host/sdk/components/StorageServer/server/src/storageServer_rpc.c:504: storage
medium size: 2097152 bytes
INFO: /host/sdk/components/StorageServer/server/src/storageServer_rpc.c:522: client 1:
offset=0, size=1048576
INFO: /host/sdk/components/StorageServer/server/src/storageServer_rpc.c:522: client 2:
offset=1048576, size=1048576
INFO: /host/sdk/libs/os_filesystem/src/OS_FileSystem.c:125: Maximizing file system
according to size reported by the storage layer (1048576 bytes)
INFO: /host/sdk/libs/os_filesystem/src/lib/SpifFs.c:182: Using SPIFFS (phys_addr = 0,
phys_size = 1048576, phys_erase_block = 4096, log_block_size = 4096, log_page_size = 256)
INFO: /host/sdk/libs/os_filesystem/src/OS_FileSystem.c:125: Maximizing file system
according to size reported by the storage layer (1048576 bytes)
INFO: /host/sdk/libs/os_filesystem/src/lib/SpifFs.c:182: Using SPIFFS (phys_addr = 0,
phys_size = 1048576, phys_erase_block = 4096, log_block_size = 4096, log_page_size = 256)
INFO: /host/sdk/demos/demo_training_course/src/components/TestApp/src/TestApp.c:135:
Demo completed successfully.
```

Code Block 2: RPi 3 B+ Terminal

Congratulations, you have successfully started your first TRENTOS-M application on a SBC.



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